



Lecture 8

Robustness to Non-linear Effect Heterogeneity

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Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation in non-linear GLMs is appropriately specified.

Why might this assumption fail?

- The nature of the conditional relationship is not appropriately captured by the model's form.
- The model imposes conditionality that is not really there.

Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation in non-linear GLMs is appropriately specified.

What kinds of inferential instability can result?

- Misleading conclusions about the average effect.
- Improper estimation of effect sizes/directions (e.g., Simpson's paradox).
- Poor predictions.

Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation in non-linear GLMs is appropriately specified.

What tools exist to evaluate robustness?

- Interflex models
- Smooth interactions with GAMs

Today's Robustness Question

What assumption/abstraction are we perturbing?

- That the nature of the relationship between x and y regarding moderation in non-linear GLMs is appropriately specified.

What are the limits of these tools?

- In essence, all variables are moderators - which to investigate?
- Evidence for moderation is often low power.
 - Interaction effects are typically estimated much less precisely than average effects.
 - Reliable estimation requires sufficient support across combinations of x and the moderator.

Problems With Linear Interactions

Consider the following model:

$$y = a + b_1X + b_2D + b_3X \times D + e$$

Hainmueller, Mummolo and Xu argue that our linear interaction models like above suffer from two potential problems:

- The conditional partial effect of each variable is a linear function of the other: $b_{D|X} = b_2 + b_3X$. Thus for every additional unit of X the effect of D changes by the same amount.
- Support - most political scientists do not pay attention to the joint density of the interacting variables and what this implies about our ability to make inferences.

Linearity

Consider the same model as before

$$y = b_0 + b_1X + b_2D + b_3X \times D + e$$

Consider two different values of $D : \{d_1, d_2\}$, the effect of this contrast is:

$$\begin{aligned}\hat{Y}(D = d_1|X) - \hat{Y}(D = d_2|X) &= (a + b_1X + b_2d_1 + b_3Xd_1) \\ &\quad - (a + b_1X + b_2d_2 + b_3Xd_2) \\ &= b_2(d_1 - d_2) + b_3X(d_1 - d_2)\end{aligned}$$

The effect will only be appropriately estimated if *both* functions of X (for d_1 and d_2 are linear).

Support

$$b_2(d_1 - d_2) + b_3X(d_1 - d_2)$$

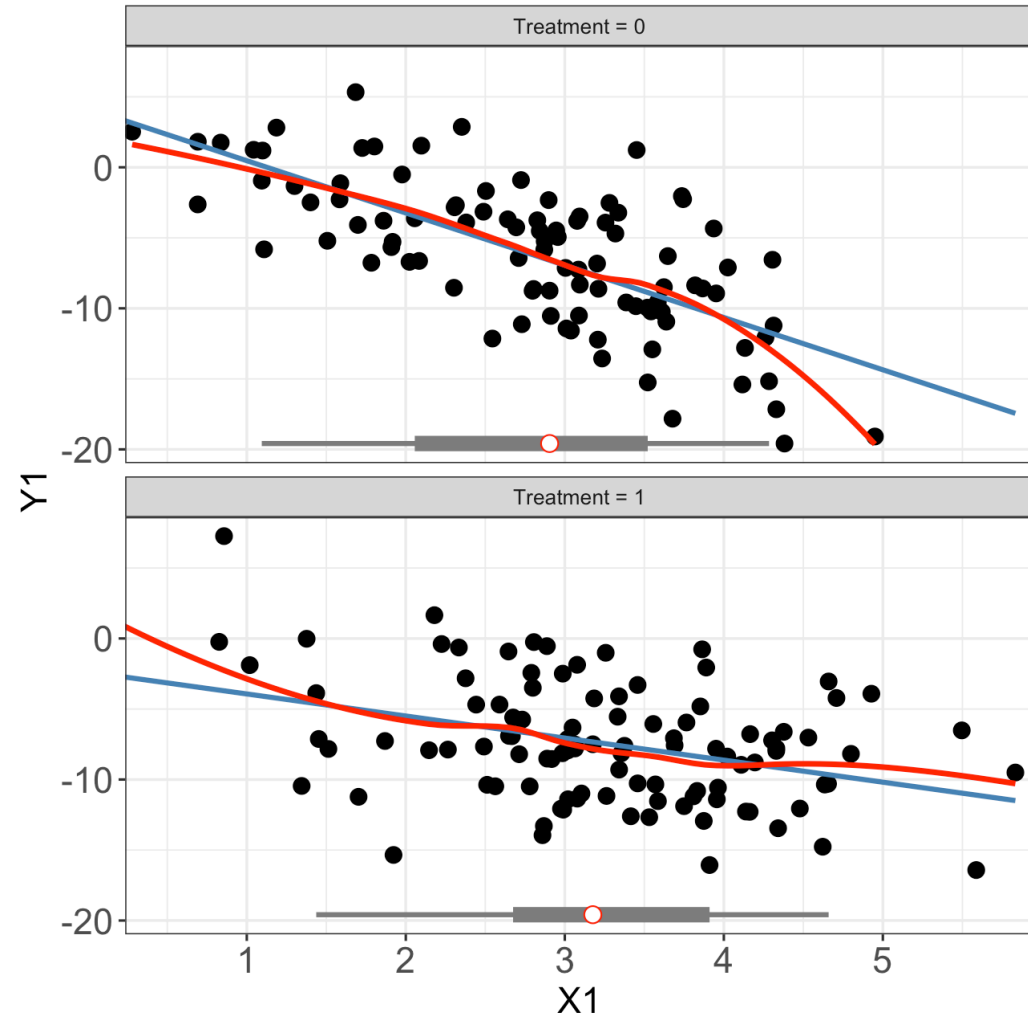
The equation above makes it clear that to reliably estimate the treatment:

- We must have reasonable data density at both d_1 and d_2 for each value of X .
- Failure of this condition results in interaction effects that are a function of interpolation and extrapolation.

Diagnostics I

```
library(interflex)
set.seed(43901)
X1 <- rnorm(200, 3, 1)
X2 <- runif(200, -3, 3)
e <- rnorm(200, 0, 4)
D1 <- rbinom(200, 1, .5)
Y1 <- 5-4*X1 -9*D1 + 3*D1*X1 + e
Y2 <- 2.5- X2^2 -5*D1 + 2*D1*X2^2 + e
dat <- as.data.frame(cbind(Y1,Y2,D1,X1, X2))
dat$D10 <- 1-dat$D1
i1 <- interflex(estimator="raw",
               dat,
               "Y1",
               "D1",
               "X1",
               treat.type="discrete")
plot(i1)
```

Baseline group not specified; choose treat = 0 as the baseline group.

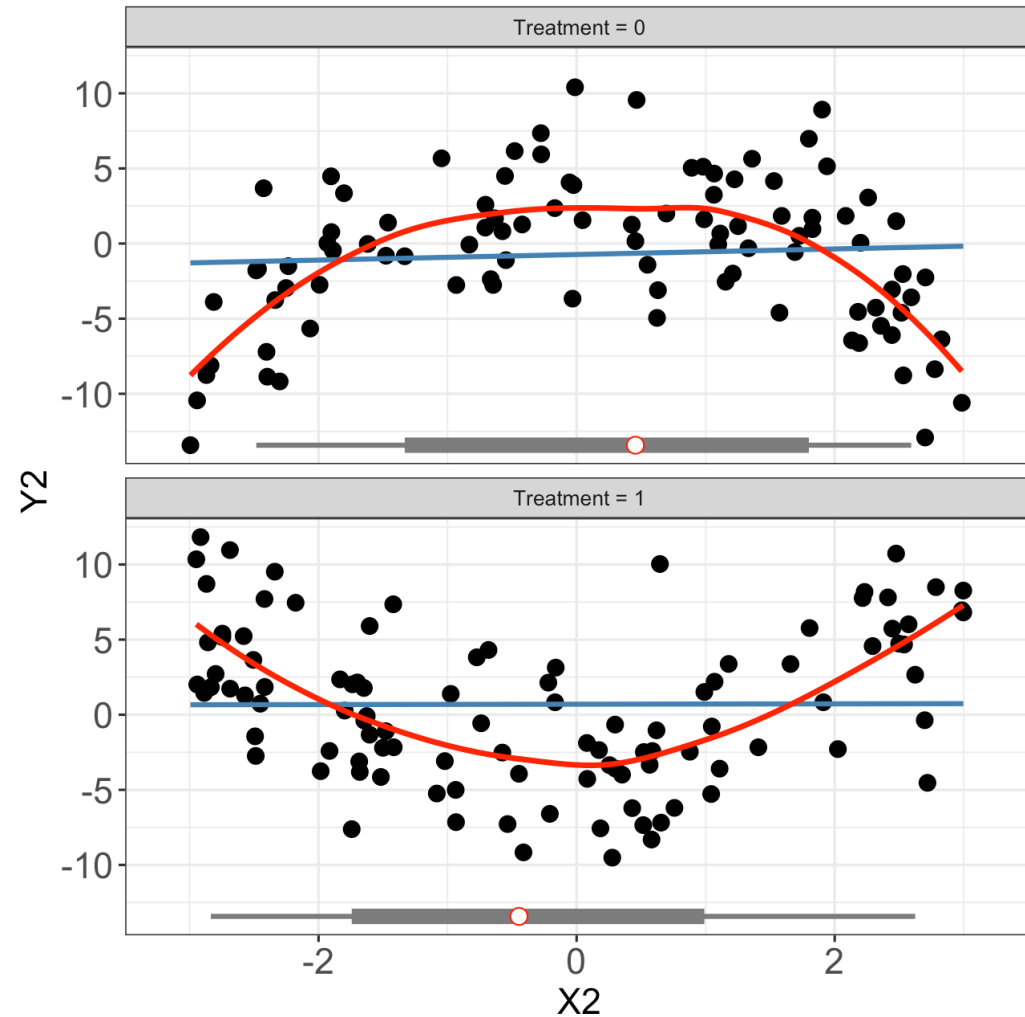


Diagnostics II

```
library(interflex)
i2 <- interflex(estimator="raw",
               dat,
               "Y2",
               "D1",
               "X2",
               treat.type="discrete")

plot(i2)
```

Baseline group not specified; choose treat = 0 as the baseline group.



Binning Estimator for Interaction Effects

- Discretize X using its three terciles.

$$G_1 = \begin{cases} 1, & \text{if } X < \delta_{\frac{1}{3}} \\ 0, & \text{Otherwise} \end{cases} \quad G_2 = \begin{cases} 1, & \text{if } \delta_{\frac{1}{3}} \geq X < \delta_{\frac{2}{3}} \\ 0, & \text{Otherwise} \end{cases} \quad G_3 = \begin{cases} 1, & \text{if } \delta_{\frac{2}{3}} \geq X \\ 0, & \text{Otherwise} \end{cases}$$

- Pick an evaluation point, x_j , in each of the J bins (usually the median of the x values within the bin)
- Estimate the model:

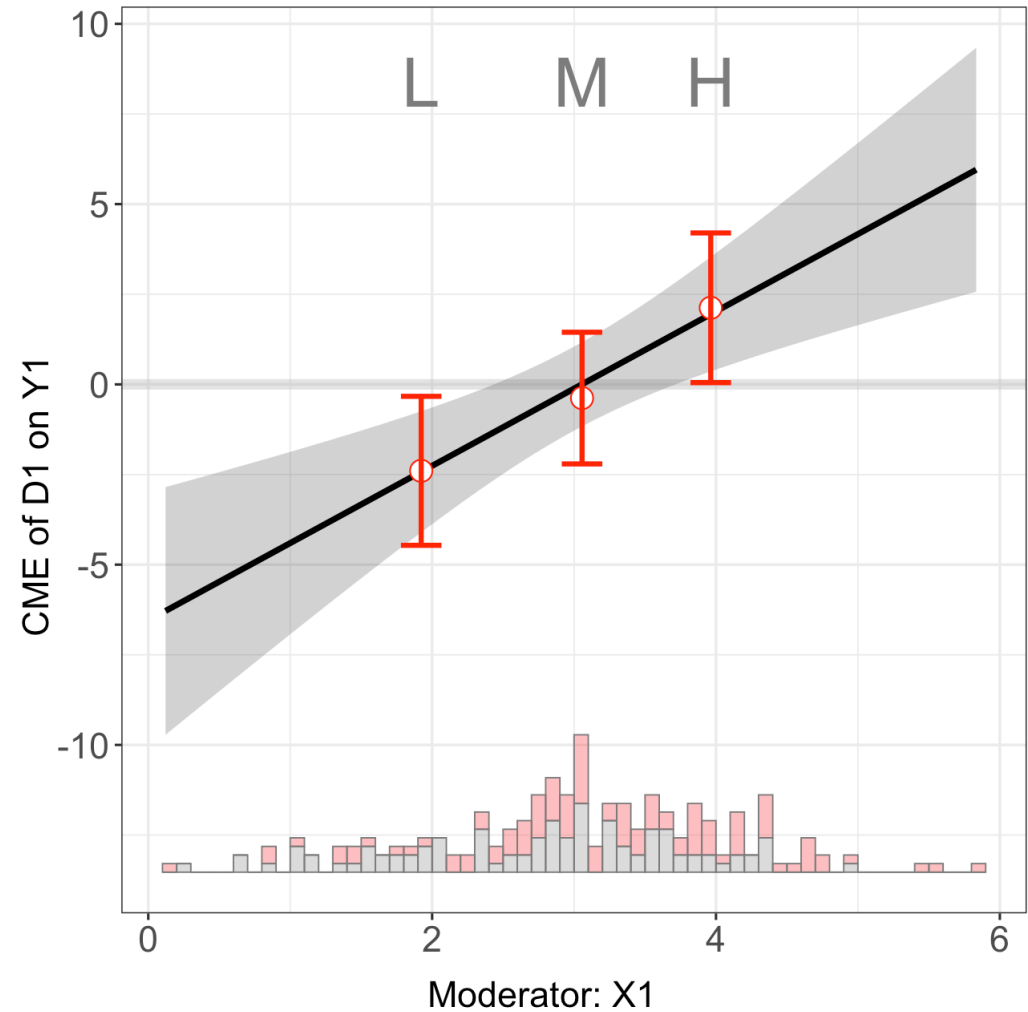
$$Y = \sum_{j=1}^J \{u_j + \alpha_j D + \eta_j(X - x_j) + \beta_j(X - x_j)D\} G_j + \dots + \varepsilon$$

Advantages over Linear Interaction Effect

- More flexible for non-linear functional forms. It fits separate interactions to each bin.
- Since $(X - x_j) = 0$ at the evaluation point, the partial conditional effect within each bin is just α_j .
- Binning ensures that there is sufficient joint support on X and D since they are constructed from X .
- This model nests the linear interaction model, so it can serve as a test of the linear interaction effect model.

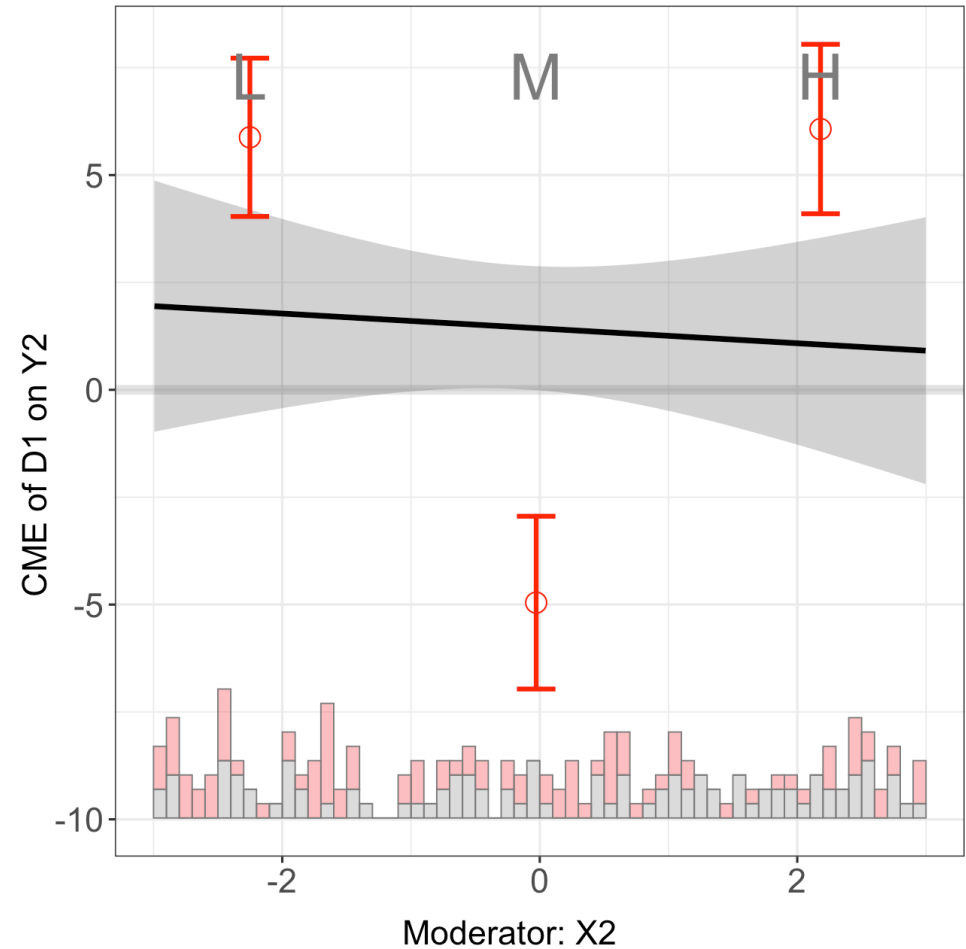
Binning Estimator

```
b1 <- interflex(estimator="binning",  
               dat,  
               "Y1",  
               "D1",  
               "X1",  
               treat.type="discrete")  
  
plot(b1)
```



Binning Estimator II

```
b2 <- interflex(estimator="binning",  
               dat,  
               "Y2",  
               "D1",  
               "X2",  
               treat.type="discrete")  
  
plot(b2)
```



Kernel Estimator

A kernel estimator is more flexible allowing the conditional partial effect to change across the values of X .

$$Y = f(X) + g(X)D + \gamma(X)Z + \varepsilon$$

- This is the standard linear interaction effect model when $f(x) = \mu + \nu X$ and $g(x) = \alpha + \beta X$ and $\gamma(X) = \gamma$.

Figure

```
k1 <- interflex(estimator="kernel",  
               dat,  
               "Y1",  
               "D1",  
               "X1",  
               treat.type="discrete",  
               parallel=FALSE)
```

```
## Baseline group not specified; choose treat  
## Cross-validating bandwidth ...
```

```
## Optimal bw=0.6184.  
## Number of evaluation points:50
```

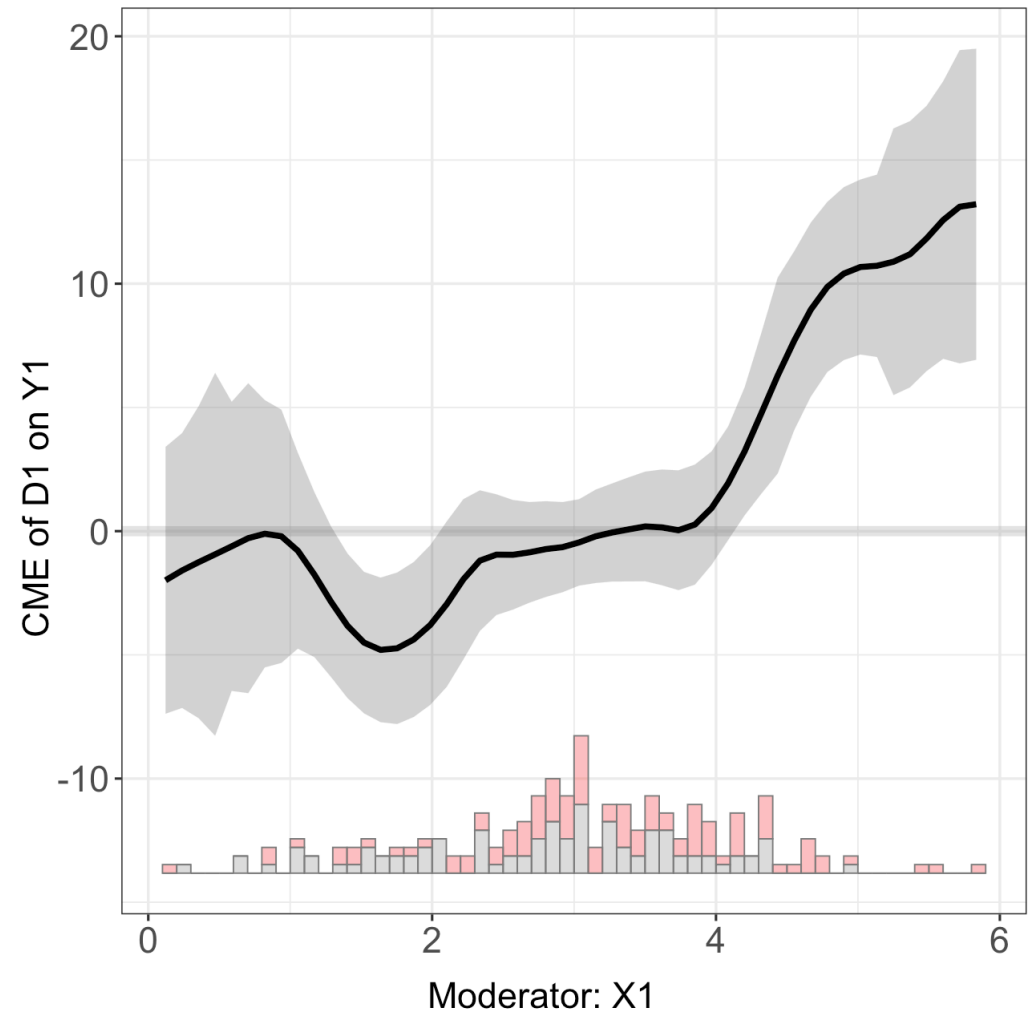
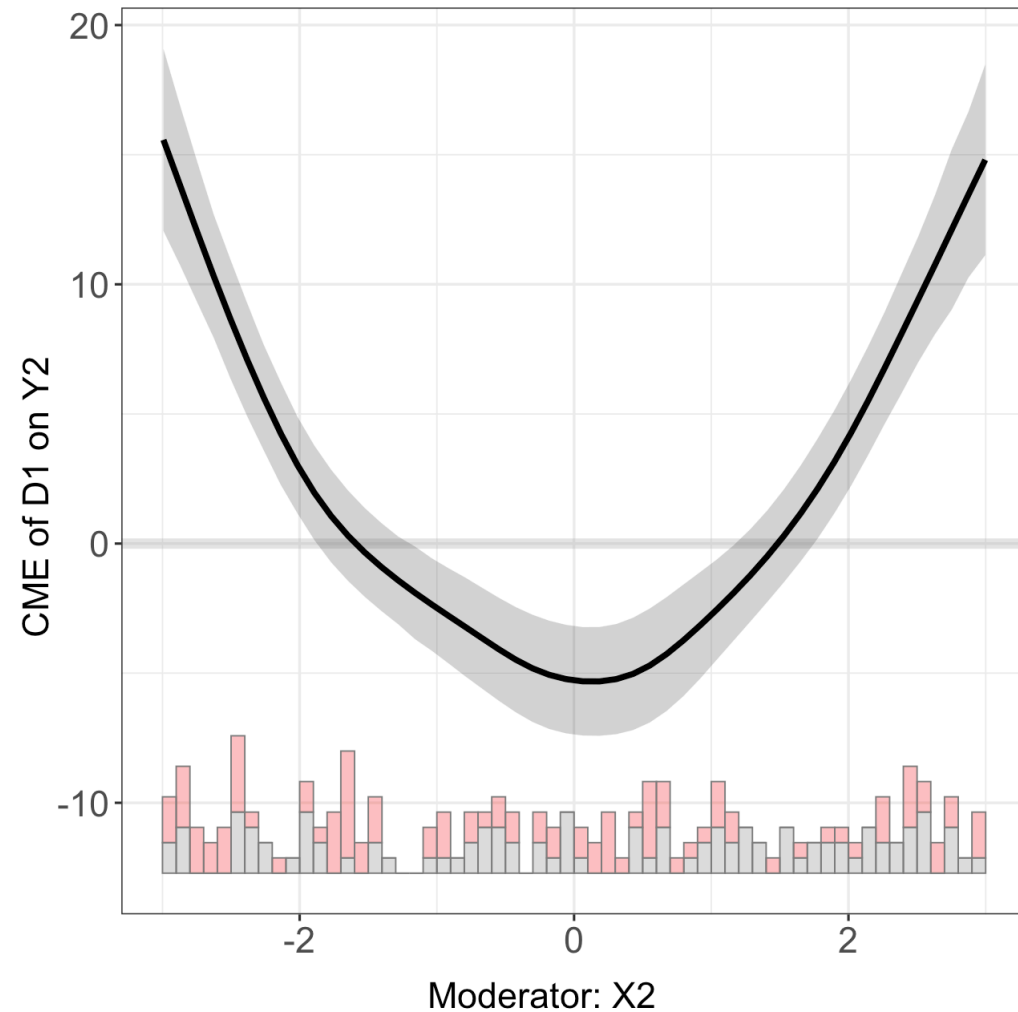


Figure II

```
k2 <- interflex(estimator="kernel",  
               dat,  
               "Y2",  
               "D1",  
               "X2",  
               treat.type="discrete",  
               parallel=FALSE)
```

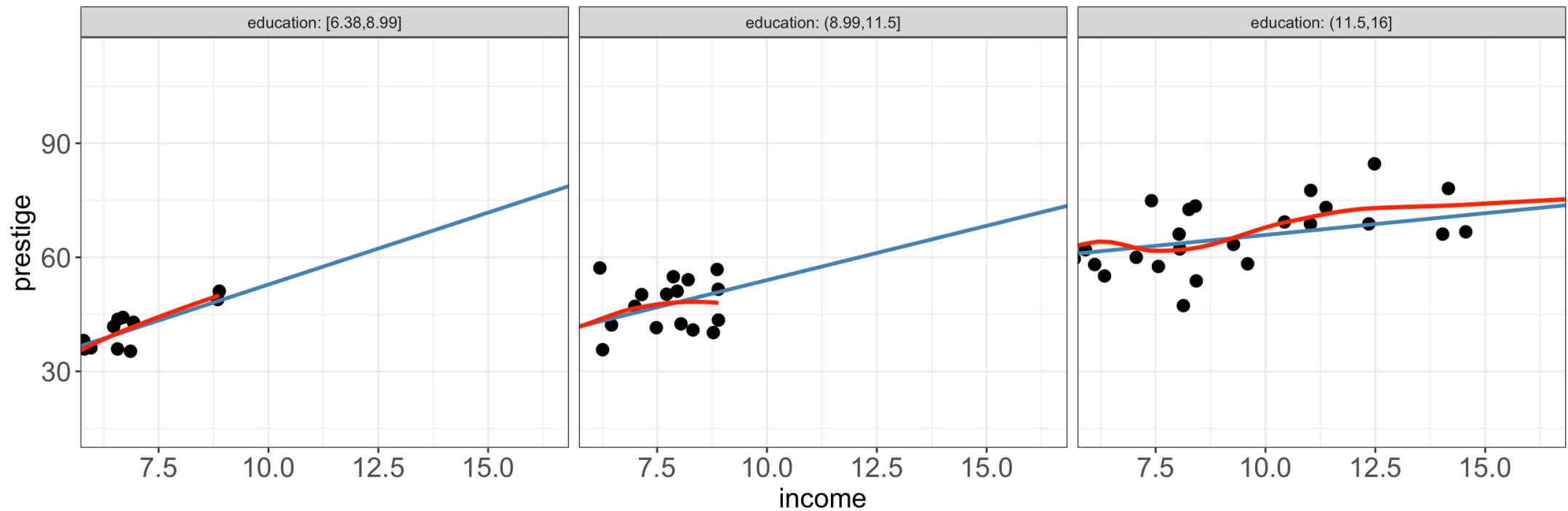
```
## Baseline group not specified; choose treat  
## Cross-validating bandwidth ...
```

```
## Optimal bw=0.891.  
## Number of evaluation points:50
```



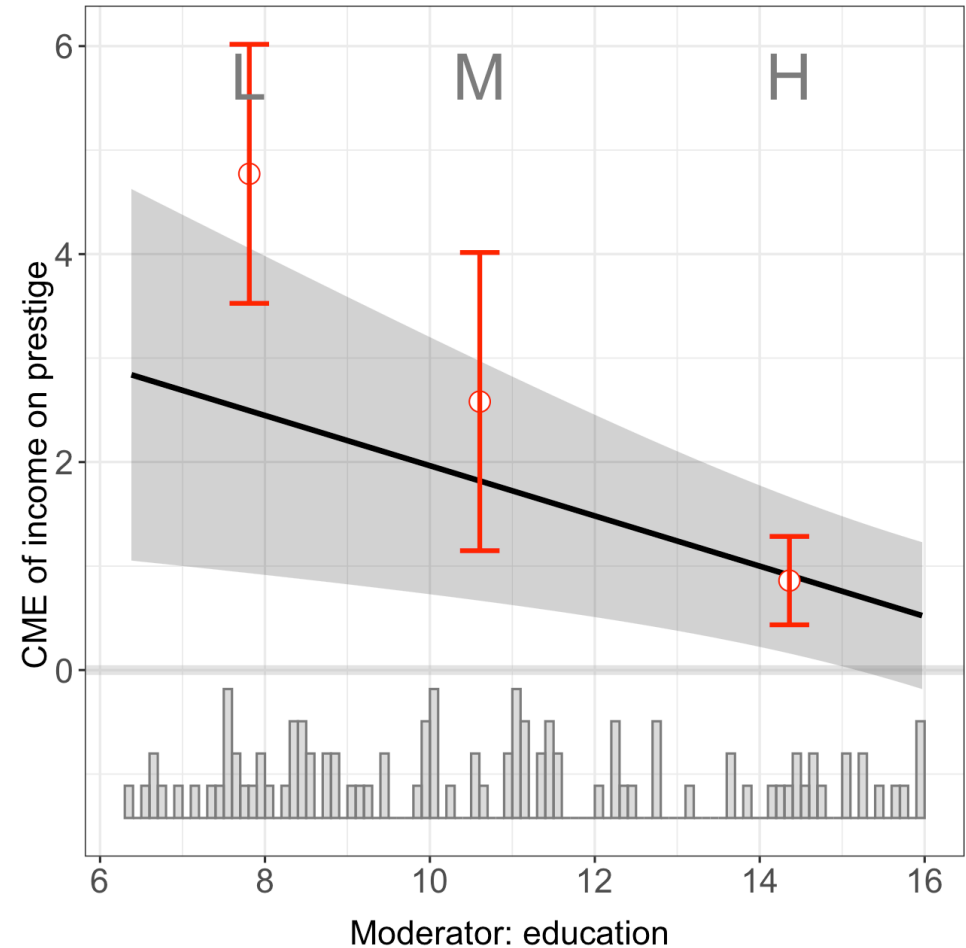
On Prestige Data

```
data(Prestige, package='carData')
Prestige$income <- Prestige$income/1000
interflex("raw", Prestige, "prestige", "income", "education", ncols=3)
```



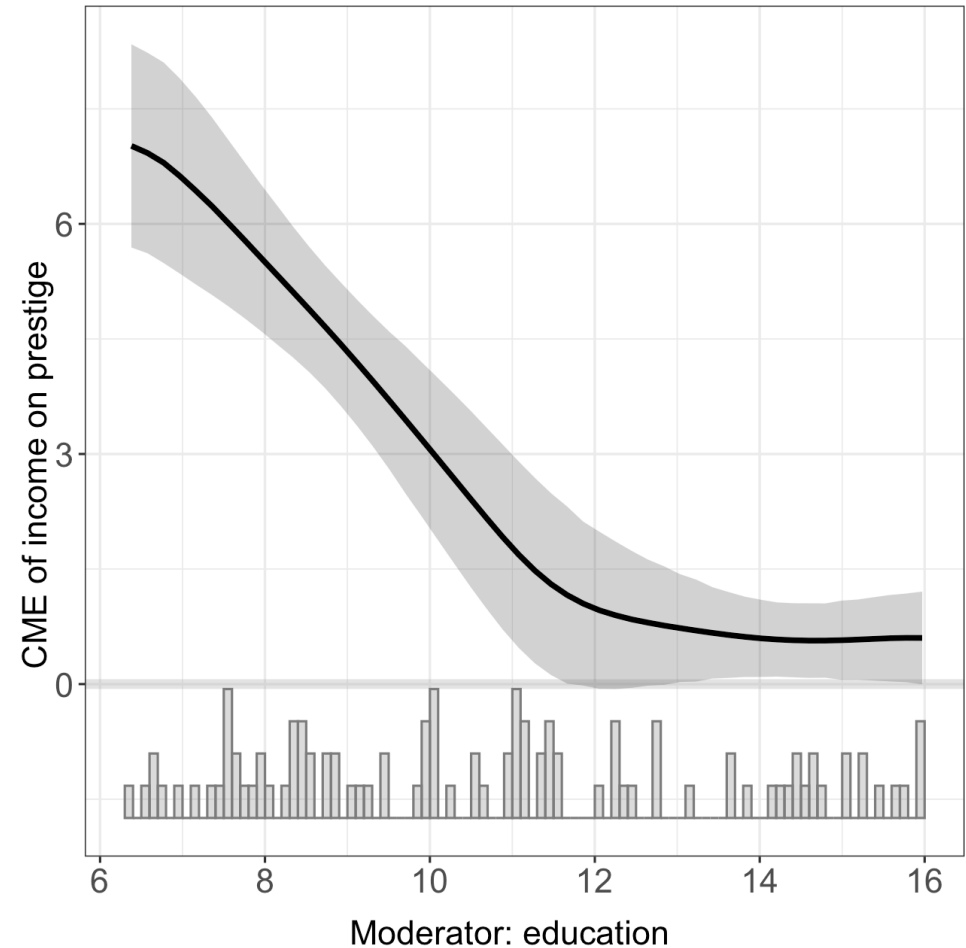
Binning Estimator: Prestige Data

```
pres.b <- interflex("binning",  
  Prestige,  
  "prestige",  
  "income",  
  "education",  
  Z = c("type",  
        "women"),  
  na.rm=T,  
  figure=FALSE)
```



Kernel Estimator: Prestige Data

```
pres.k <- interflex("kernel",  
  Prestige,  
  "prestige",  
  "income",  
  "education",  
  Z=c("type",  
      "women"),  
  na.rm=T,  
  figure=FALSE,  
  treat.type="continuous",  
  parallel=FALSE)
```



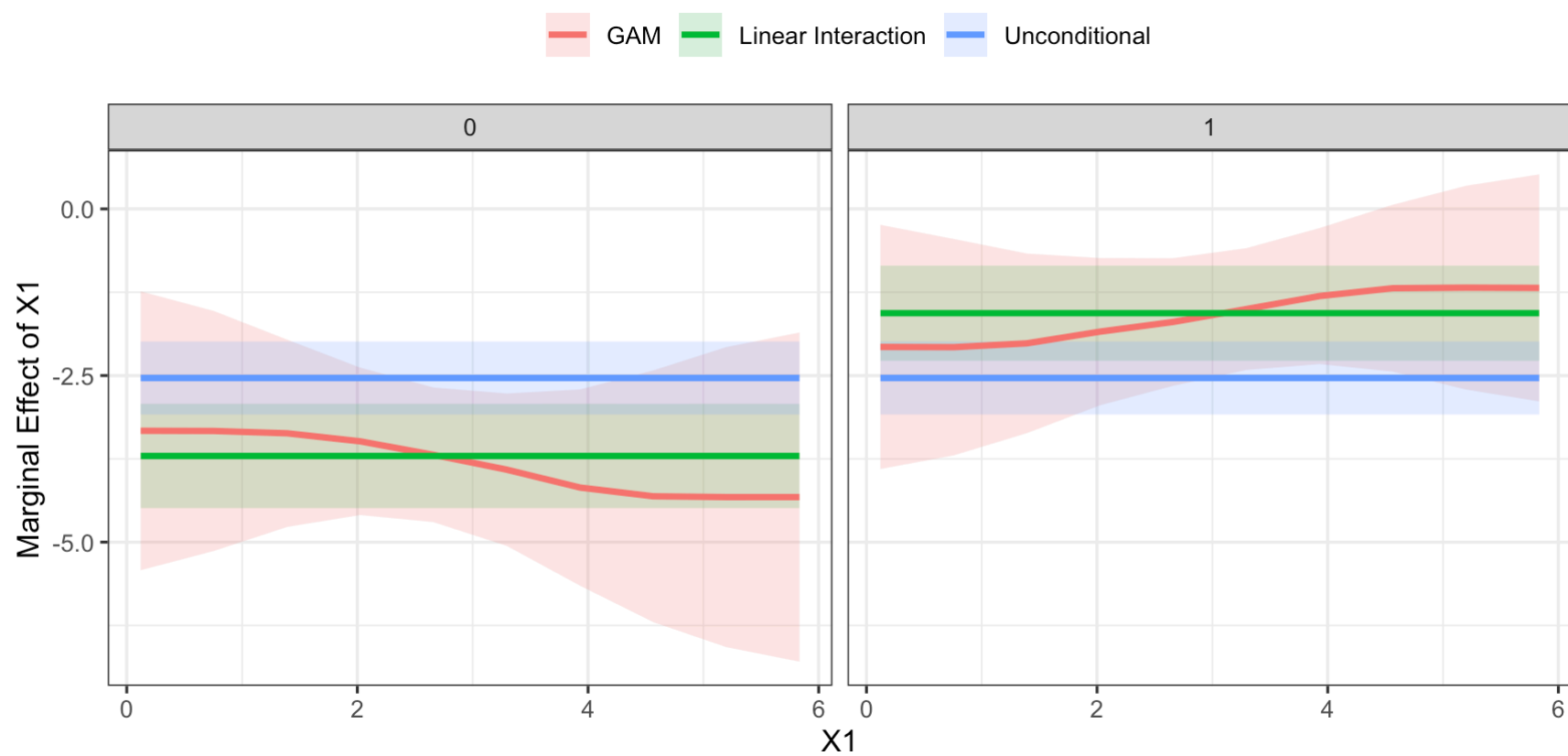


GAMs for the Fake Data I

```
library(mgcv)
dat$D1 <- as.factor(dat$D1)
mod0u <- gam(Y1 ~ X1 + D1, data=dat)
mod0 <- gam(Y1 ~ X1*D1, data=dat)
mod1 <- gam(Y1 ~ s(X1, by=D1) + D1,
            data=dat)
anova(mod0, mod1, test="F")
```

```
## Analysis of Deviance Table
##
## Model 1: Y1 ~ X1 * D1
## Model 2: Y1 ~ s(X1, by = D1) + D1
##   Resid. Df Resid. Dev      Df Deviance      F Pr(>F)
## 1    196.00    2947.9
## 2    194.36    2910.8 1.6418   37.127 1.5153 0.2242
```

Conditional Effects of X1



Robustness of Conditional Effect of X1

```
## # A tibble: 20 × 5
## # Rowwise:
##       X1 D1      ovl_u_li ovl_li_g ovl_u_g
##   <dbl> <fct>      <dbl>      <dbl>    <dbl>
## 1 0.121 0        0.0829      0.532    0.347
## 2 0.121 1        0.129       0.511    0.432
## 3 0.756 0        0.0829      0.578    0.358
## 4 0.756 1        0.129       0.534    0.459
## 5 1.39 0        0.0829      0.657    0.339
## 6 1.39 1        0.129       0.585    0.470
## 7 2.03 0        0.0829      0.769    0.239
## 8 2.03 1        0.129       0.705    0.372
## 9 2.66 0        0.0829      0.877    0.138
## 10 2.66 1        0.129       0.822    0.259
## 11 3.29 0        0.0829      0.768    0.102
## 12 3.29 1        0.129       0.871    0.158
## 13 3.93 0        0.0830      0.595    0.0967
## 14 3.93 1        0.129       0.731    0.117
## 15 4.56 0        0.0829      0.511    0.124
## 16 4.56 1        0.129       0.632    0.129
## 17 5.20 0        0.0829      0.474    0.155
## 18 5.20 1        0.129       0.587    0.171
## 19 5.83 0        0.0829      0.452    0.169
## 20 5.83 1        0.129       0.559    0.195
```



Simulation

```
sim_function <- function(base, alt, vbl, byvar, m_num = 10, s_len = 10,
                        type = c("slopes", "comparisons", "predictions",
                                "avg_slopes", "avg_comparisons", "avg_predictions"),
                        R = 25, progress=TRUE,
                        ...){
  type_fun <- match.arg(type)
  nlevs <- function(x){
    if(inherits(x, "factor"))length(levels(x))
    else length(unique(na.omit(x)))
  }
  tmp <- insight::get_data(base)
  sim_resp <- try(simulate(base)[[1]])
  if(inherits(sim_resp, "try-error")){
    stop("simulate() failed on base model.\n")
  }else{
    setvars <- union(vbl, byvar)
    vars <- lapply(setvars, \(x){
      nl <- nlevs(tmp[[x]])
      if(nl > m_num){
        seq_range(tmp[[x]], n=s_len)
      }else{
        if(inherits(tmp[[x]], "factor"))levels(tmp[[x]])
        else sort(unique(na.omit(tmp[[x]])))
      }
    })
    names(vars) <- setvars
    vars$model <- base
    vars$grid_type = "mean_or_mode"
    dg <- do.call(marginaleffects::datagrid, vars)
```

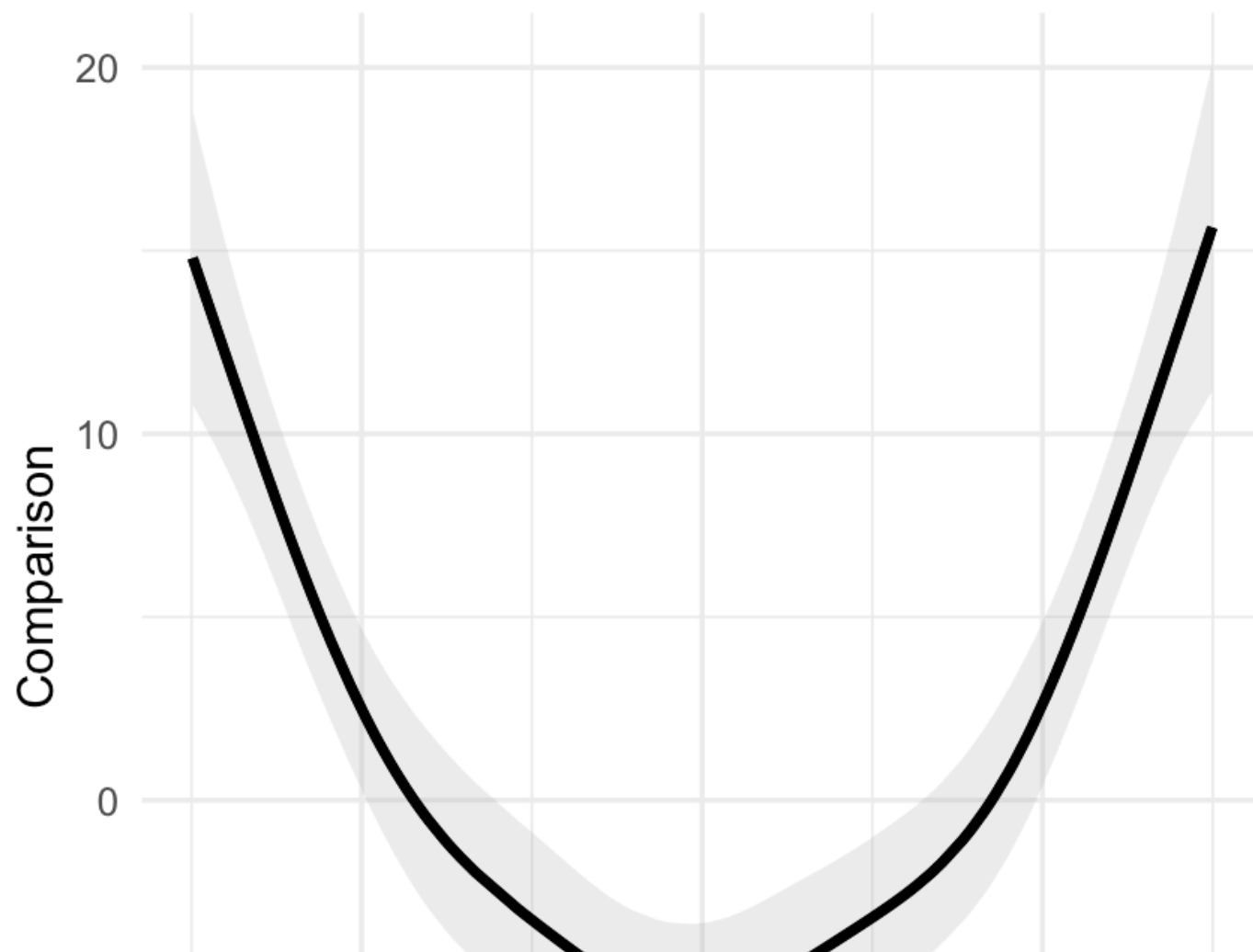



GAMs for the Fake Data II

```
mod0b <- gam(Y2 ~ X2*D1, data=dat)
mod1b <- gam(Y2 ~ s(X2, by=D1) + D1,
             data=dat)
anova(mod0b, mod1b)
```

```
## Analysis of Deviance Table
##
## Model 1: Y2 ~ X2 * D1
## Model 2: Y2 ~ s(X2, by = D1) + D1
##   Resid. Df Resid. Dev      Df Deviance      F      Pr(>F)
## 1      196.00      5080.2
## 2      188.02      2781.8 7.9818    2298.4 19.651 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Treatment Effect Plot



GAMs for the Prestige Data

```
library(mgcv)
Prestige <- na.omit(Prestige)
mod_int0 <- gam(prestige ~ log(income)*education +
  women + type, data=Prestige)
mod_int1 <- gam(prestige ~ ti(income) + ti(education) +
  ti(income, education) + women + type,
  data=Prestige)
anova(mod_int0, mod_int1)
```

```
## Analysis of Deviance Table
##
## Model 1: prestige ~ log(income) * education + women + type
## Model 2: prestige ~ ti(income) + ti(education) + ti(income, education) +
##      women + type
##      Resid. Df Resid. Dev      Df Deviance      F Pr(>F)
## 1      91.000      3649.6
## 2      86.746      3450.9 4.254    198.69 1.1898 0.3212
```

Interactions Conclusion

The GAM framework provides us the ability to:

- model general non-linear interactions in both continuous and discrete variables
- do most everything suggested by Hainmueller et al.